Exercise 4 – CPU Scheduling

Questions are taken from Stallings, Operating Systems Internals and Design Principles, fifth edition and Silberschatz et al., Operating System Concepts, seventh edition.

1 – Silberschatz 5.4

Consider the following set of processes, with the length of the CPU burst given in milliseconds.

Process	Burst Time	Priority
P_1	10	3
P_2	1	1
P_3	2	3
P_4	1	4
P_5	5	2

The processes are assumed to have arrived in the order P_1, P_2, P_3, P_4, P_5 all at time 0.

a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, non-preemptive priority (a smaller priority number implies a higher priority), and RR (quantum = 1).

b) What is the turnaround time of each process for each of the scheduling algorithms in part a?

c) What is the waiting time of each process for each of the scheduling algorithms in part a?

d) Which of the algorithms om part a results in the minimum average waiting time (over all processes)?

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Consider the following set of processes, with the length of the CPU burst and arrival time given in milliseconds.

Process	Burst Time	Arrival time
P_1	8	0
P_2	4	0.4
P_3	1	1

a) Draw four Gantt charts that illustrate the execution of these processes using the following scheduling algorithms: FCFS, SJF, Clairvoyant SJF (the algorithm can look into the future and wait for a shorter process that will arrive).

b) What is the turnaround time of each process for each of the scheduling algorithms in part a?

c) What is the waiting time of each process for each of the scheduling algorithms in part a?

d) Which of the algorithms om part a results in the minimum average waiting time (over all processes)?